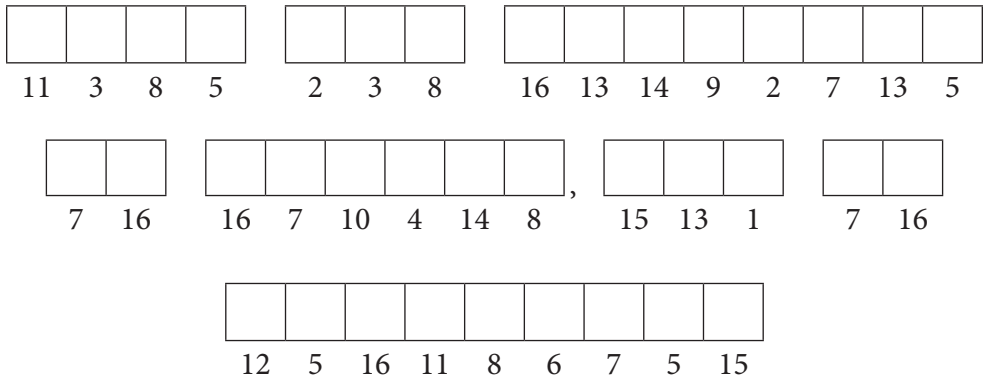
Name: _____

WORKSHEET

Solving algebraic equations



Albert Einstein

Questions

Solve the following algebraic equations for x.

1 $3x - 15 = 0$

2 $2x + 7 = 19$

3 $5x + 12 = 6 + 3x$

4 $x^2 + 4x - 5 = 0$

5 $x^2 + 10x + 21 = 0$

6 $x^2 - 7x + 12 = 0$

7 $5x - 1 = 3x + 3$

8 $2x^2 + 4x - 30 = 0$

9 $\dfrac{x^2 - 6}{x} = 1$

10 $\dfrac{7x^2 + 2x - 3}{x} = 3(x + 2)$

11 $3x^2 + 5x + 7 = 2x^2 + 8x + 11$

12 $5x^2 + 3x - 4 = 14x + 8$

13 $\dfrac{x - 6}{4} = \dfrac{x - 4}{x}$

14 $7x^2 + 10x - 4 = x^2 - 15x + 5$

15 $\dfrac{x}{x^2 + 3} = \dfrac{2}{x + 5}$

16 $\dfrac{2}{x - 2} = \dfrac{x - 1}{x^2 - 4}$

Solutions

A $x = -\dfrac{4}{5}, 3$

D $x = 5$

E $x = -5, 3$

G $x = 2, 3$

H $x = -3$

I $x = 2$

L $x = -4\dfrac{1}{2}, \dfrac{1}{3}$

M $x = -\dfrac{1}{2}, 1\dfrac{1}{2}$

N $x = -7, -3$

O $x = 2, 8$

P $x = -5, 1$

R $x = 3, 4$

S $x = -5, 2$

T $x = 6$

U $x = -2, 3$

W $x = -1, 4$

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